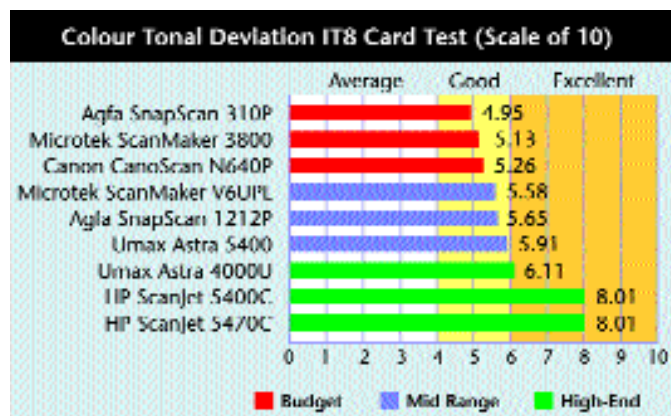




the Microtek ScanMaker 3800 could go beyond 5—they scored 5.26 and 5.13 respectively.

The mid range segment proved to be a mixed bag. While the Mustek 1200ED and the Microtek ScanMaker X6EL scored below 18 in the greyscale deviation test (17 and 16 respectively), the rest detected more than 18 bars. The Agfa SnapScan 1212P, Microtek ScanMaker V6UPL and Umax Astra 4400 scored 21 each. These three scanners did well in the colour deviation test too, scoring over 5 points. Surprisingly, the Umax Astra 5400 took top honours in the colour deviation test, scoring a healthy 5.91. The Astra 5400 has been quite an underperforming scanner, with its younger sibling, the Umax Astra 4400, trouncing it in almost all the performance tests.

All the scanners in the high-end segment detected over 20 bars in the greyscale tonal deviation test. The HP ScanJet 5470C and the Microtek ScanMaker X12USL stood out by detecting 21 bars each. But the colour deviation test was where these big boys

got to show their real muscle. Scoring 5 was not a problem for these scanners, but the real champions were the HP ScanJet series of scanners—the 5400C and the 5470C scored an impressive 8.01 each. A scanner that can replicate such sublime variations in tone as seen on an IT8 card speaks a lot about the quality of its scan engine.

Colour photo scan quality: This is a test where bit depth matters the most. You need a minimum bit depth of 24 bits for decent image quality. A certain loss or distortion of information is inherent in the scanning process. This is commonly referred to as noise or artefacting. In reality, artefacting cuts a standard 24-bit depth to around 16, which directly decreases the number of colours represented and hence the quality of the scanned image suffers.

The technically inclined may wonder that a monitor can display only 24 bits of colour data, so what happens to a 48-bit scan? Well, the answer is simple: with more information in each pixel, the image has smoother colour gradations and a much improved overall image colour reproduction.

The quality of the scanned photo in the budget scanners was pretty average. This could be directly attributed to the lower resolutions that these scanners support. A score of 5 out of 10 should be considered fairly average. The highlights and shadows of the photo were more subdued and one could distinctly observe jagged edges in parts of the image. Artefacting was also fairly pronounced—we found about five to eight white patches (artefacts) with these scanners. The best scores were logged by the Agfa SnapScan 310P and the Mustek BearPaw 1200CU (6.79 points each). Their scanned images had almost zero artefacting, the colours were smooth and the edges sharp. The Agfa SnapScan 310P scores are quite astonishing—this scanner has



Scanner Calibration

Colour calibrating your scanner can go a long way towards ensuring that what you scan, what you see on screen and what you print look the same.

Basic visual calibration: The first step is to calibrate your monitor. The next step is to scan something and make adjustments using a gamma correction tool like Adobe Gamma until the colours on the scanned image on your monitor and the actual photograph match.

Use any high-quality photographic image with a wide range of tonal values. Ensure that before scanning, all auto-

matic colour correction is turned off. After scanning, adjust the controls from within your scanning software and rescan until what you scan matches your monitor display. Note all adjustments and save them as a profile for future use.

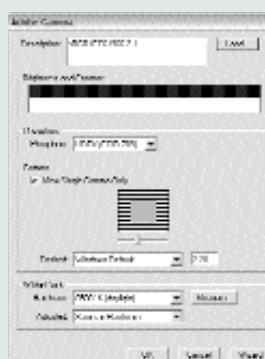
Colour calibration with ICC profiles: ICC profiles provide a more thorough way to ensure consistent colour. These files are specific to each device on your system and contain information about how that device produces colour. If your scanner comes with a pre-made colour profile for your scanner model, use it—it may give better results than manual calibration.

Colour calibration using an IT8 card: Calibration or profiling software may come with an IT8 scanner target—a printed card that includes photographic images, greyscale bars and different colour bars. Different manufacturers have their own IT8 cards, but they all generally conform to the same standard for colour representation. The IT8 card requires a digital refer-

ence file (colour profile) specific to that image. However, some scanners come with pre-defined profiles; in order to accurately calibrate your scanner you will require profiling software that comes bundled with the IT8 card.

First scan the IT8 card. The scanned image will be an RGB image; change it to Lab colour (this can be done using a calibration software such as Color Blind Matchbox). Now compare the lab values of the scanned patches on the IT8 card with the values in the digital reference file of the IT8 card. The software can now calculate your input system's colour error for each patch. The software will use this information to build a corrective colour table. This table and its supporting information can be saved as a colour profile.

Preferably, calibrate your scanner every month, depending on how much you use your scanner. If you make changes to your software or hardware, it may be necessary to re-calibrate.



Adjust gamma settings for optimal colour matching